

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

pd.set_option('display.max_columns', None)

print("Libraries loaded successfully")
```

Libraries loaded successfully

```
from google.colab import files

uploaded = files.upload()
```

superstore_...aset (1).csv
superstore_final_dataset (1).csv(text/csv) - 2124717 bytes, last modified: 25/5/2026 - 100% done
 Saving superstore_final_dataset (1).csv to superstore_final_dataset (1).csv

```
df = pd.read_csv("superstore_final_dataset (1).csv", encoding='latin1')

df.head()
```

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_
0	1	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire
1	2	CA-2017-152156	8/11/2017	11/11/2017	Second Class	CG-12520	Claire
2	3	CA-2017-138688	12/6/2017	16/06/2017	Second Class	DV-13045	Darrin Van
3	4	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Do
4	5	US-2016-108966	11/10/2016	18/10/2016	Standard Class	SO-20335	Sean O Do

Next steps:

[Generate code with df](#)

[New interactive sheet](#)

```
print("Shape:", df.shape)
```

```
print("\nData types:")  
print(df.dtypes)
```

```
print("\nMissing values:")  
print(df.isnull().sum())
```

```
Shape: (9800, 18)
```

```
Data types:
```

```
Row_ID          int64  
Order_ID        object  
Order_Date      object  
Ship_Date       object  
Ship_Mode       object  
Customer_ID     object  
Customer_Name   object  
Segment         object  
Country         object  
City            object  
State           object  
Postal_Code     float64  
Region          object  
Product_ID      object  
Category        object  
Sub_Category    object  
Product_Name    object  
Sales           float64
```

```
dtype: object
```

```
Missing values:
```

```
Row_ID          0  
Order_ID        0  
Order_Date      0  
Ship_Date       0  
Ship_Mode       0  
Customer_ID     0  
Customer_Name   0  
Segment         0  
Country         0  
City            0  
State           0  
Postal_Code     11  
Region          0  
Product_ID      0  
Category        0  
Sub_Category    0  
Product_Name    0  
Sales           0
```

```
dtype: int64
```

```
# Data cleaning
```

```
# Convert date columns
df["Order_Date"] = pd.to_datetime(df["Order_Date"], dayfirst=True)
df["Ship_Date"] = pd.to_datetime(df["Ship_Date"], dayfirst=True)

# Fill missing postal codes
df["Postal_Code"] = df["Postal_Code"].fillna(0)

# Check duplicates
print("Duplicate rows:", df.duplicated().sum())

print("\nMissing values after cleaning:")
print(df.isnull().sum())
```

Duplicate rows: 0

Missing values after cleaning:

```
Row_ID      0
Order_ID    0
Order_Date  0
Ship_Date   0
Ship_Mode   0
Customer_ID 0
Customer_Name 0
Segment     0
Country     0
City        0
State       0
Postal_Code 0
Region      0
Product_ID  0
Category    0
Sub_Category 0
Product_Name 0
Sales       0
dtype: int64
```

```
# KPI calculations
```

```
total_revenue = df["Sales"].sum()

average_order_value = df["Sales"].mean()

total_orders = df["Order_ID"].nunique()

print("Total Revenue: ", round(total_revenue,2))
print("Average Order Value: ", round(average_order_value,2))
print("Total Orders: ", total_orders)
```

```
Total Revenue: 2261536.78
Average Order Value: 230.77
Total Orders: 4922
```

```
# Sales trend over time
```

```
monthly_sales = df.groupby(df["Order_Date"].dt.to_period("M"))["Sales"].sum()
```

```

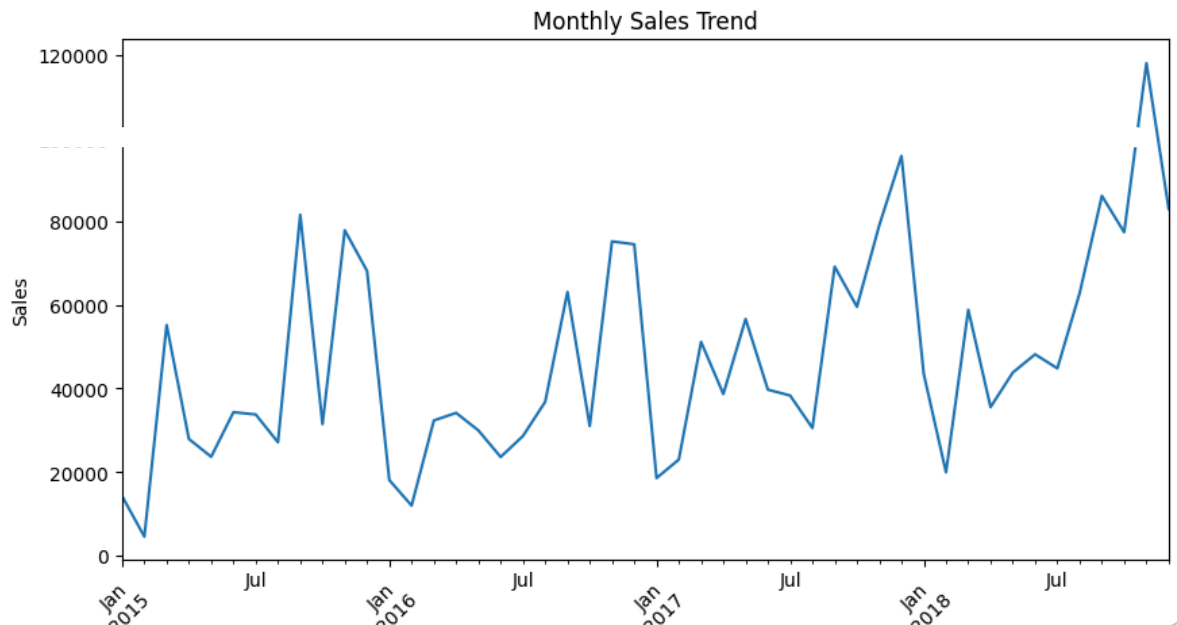
monthly_sales.plot(figsize=(10,5))

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Sales")

plt.xticks(rotation=45)

plt.show()

```



```

# Regional sales performance

region_sales = df.groupby("Region")["Sales"].sum()

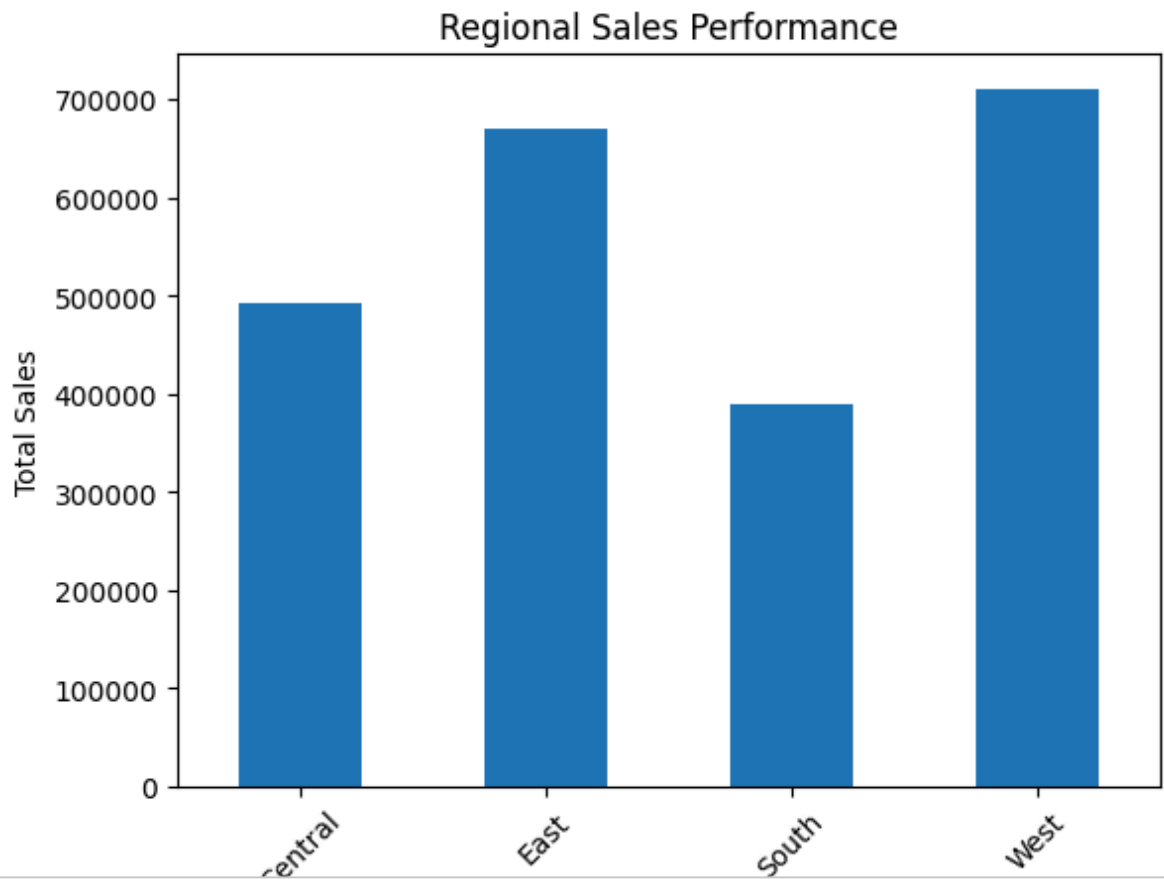
region_sales.plot(kind="bar")

plt.title("Regional Sales Performance")
plt.xlabel("Region")
plt.ylabel("Total Sales")

plt.xticks(rotation=45)

plt.show()

```



```
# Top 10 products by sales

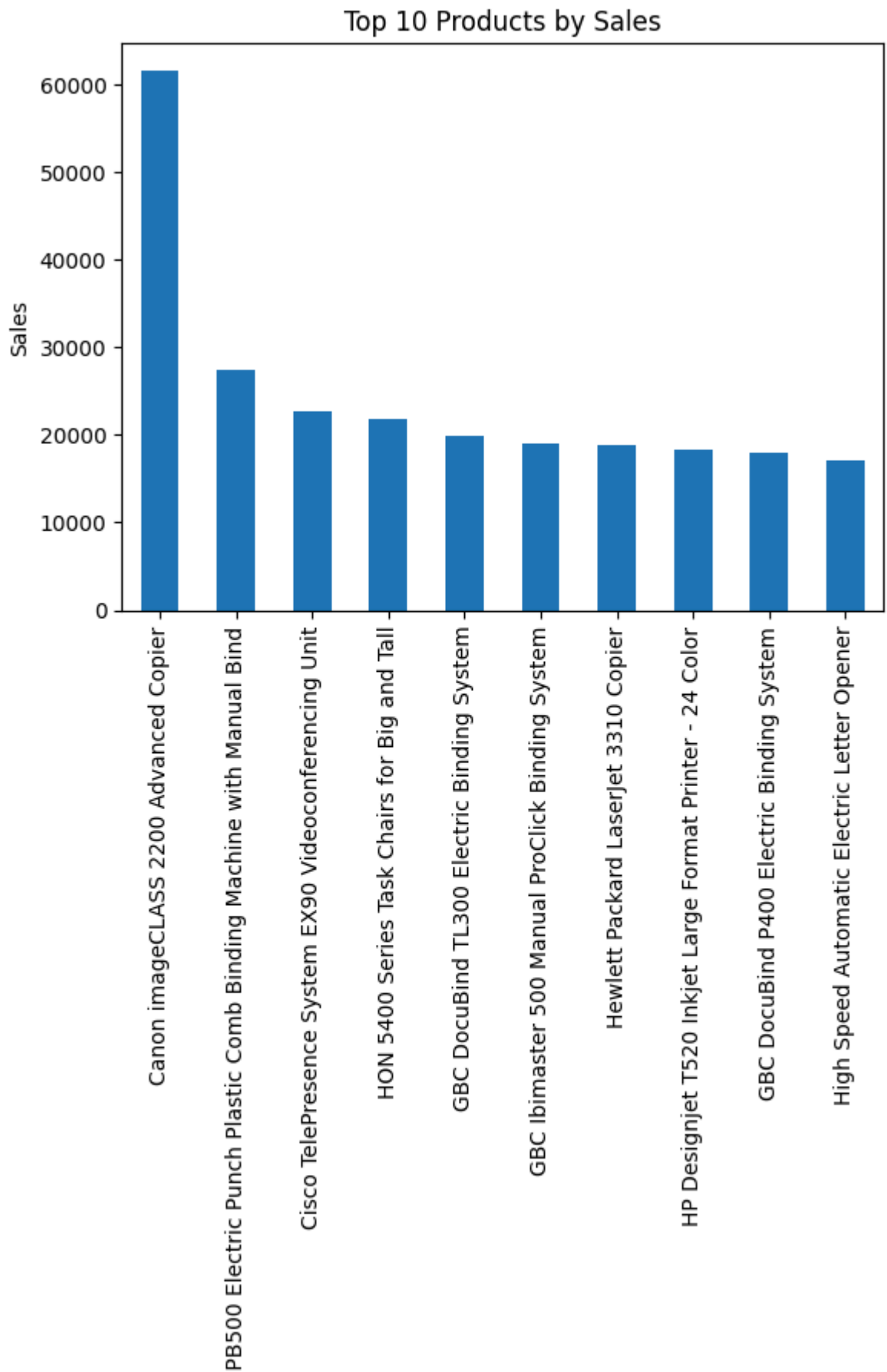
top_products = df.groupby("Product_Name")["Sales"].sum().sort_values(ascendi

top_products.plot(kind="bar")

plt.title("Top 10 Products by Sales")
plt.xlabel("Product")
plt.ylabel("Sales")

plt.xticks(rotation=90)

plt.show()
```



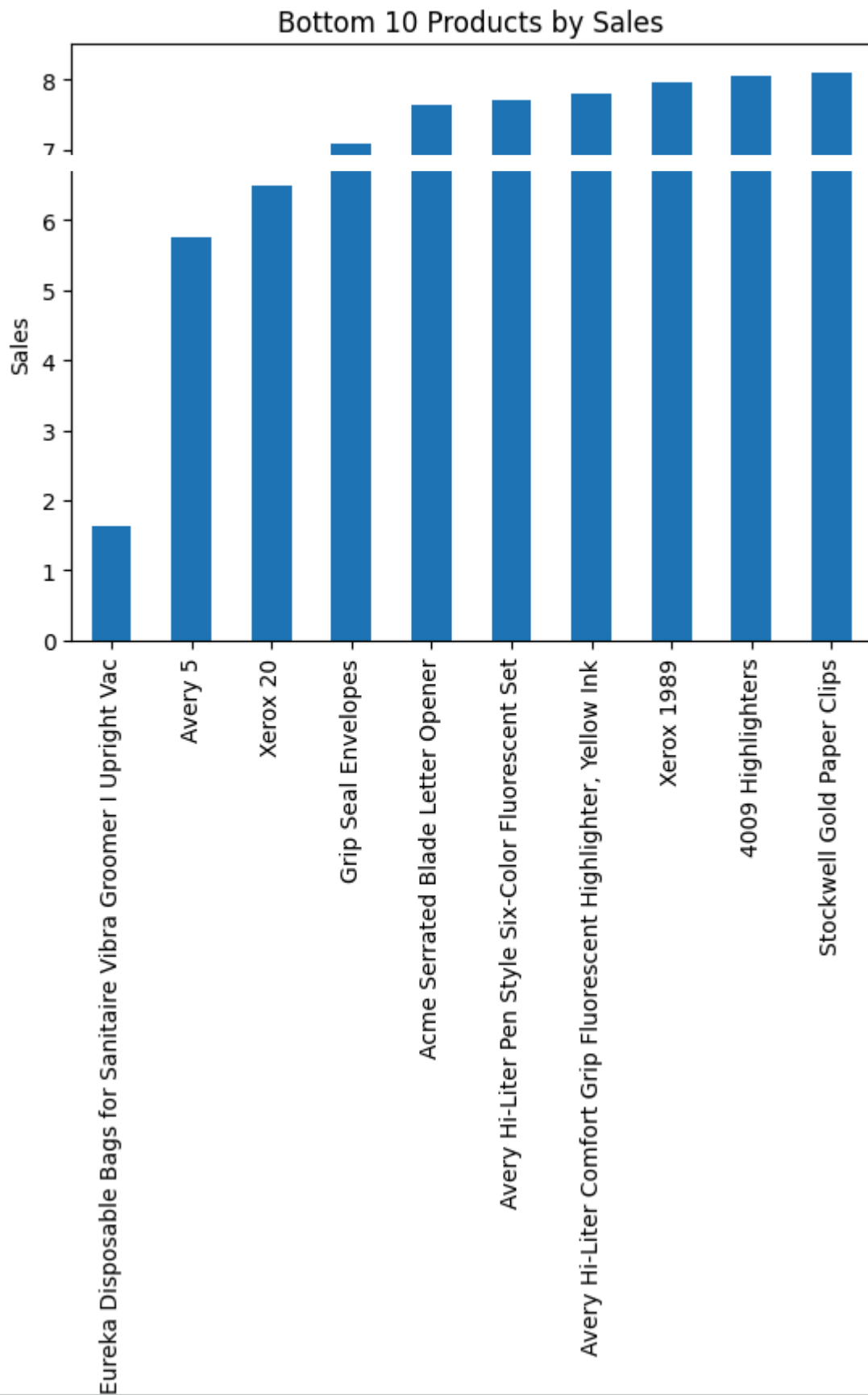
Bottom 10 products by sales

```
bottom_products = df.groupby("Product_Name")["Sales"].sum().sort_values().he  
bottom_products.plot(kind="bar")
```

```
plt.title("Bottom 10 Products by Sales")
plt.xlabel("Product")
plt.ylabel("Sales")

plt.xticks(rotation=90)

plt.show()
```



Actionable Recommendations for Alfido Tech Focus marketing efforts on high-performing regions such as West and analyze successful strategies that can be implemented in weaker regions. Improve sales performance in low-performing regions through targeted campaigns, customer engagement programs, and promotional offers. Prioritize top-selling products by maintaining sufficient inventory and increasing promotional activities. Review low-performing products and optimize pricing strategies or reduce inventory for products with consistently low sales. Use seasonal sales patterns for better demand forecasting and inventory planning.

Executive Summary

Sales analysis was conducted using the Superstore dataset to evaluate performance across regions, products, and time periods. The analysis showed an overall growth trend in sales with noticeable seasonal fluctuations. The West region generated the

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